Multi-Channel Spectrogram Fusion with Digital Image Processing Enhancement for Convolutional Neural Network-Based Speech Emotion Recognition

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***Abstract*—**Speech carries far more emotional information than its words alone, yet reliably extracting that information remains one of the harder problems in affective computing. The acoustic cues are nuanced and speaker-dependent, the mapping to discrete emotion labels is imprecise, and labelled corpora are consistently scarce. Despite this, speech emotion recognition (SER) is finding its way into clinical screening tools [2], conversational agents [1], and driver monitoring systems [3]. Most deployed solutions, however, still feed a single Mel spectrogram or MFCC representation into a convolutional network and stop there. We see two concrete problems with that choice. No single transform captures the full acoustic picture: CQT reveals harmonic structure across a log-frequency axis [30], and Chroma STFT aggregates pitch-class energy in ways that track prosodic patterns [8] — information a Mel-only model never has access to. Beyond representation breadth, the images that audio-to-image pipelines produce tend to be low-contrast, particularly in the mid-frequency bands where formant transitions encode much of the emotional signal [9],[28]. This paper addresses both problems together. We extract four spectrogram types per utterance (Mel, MFCC, CQT, Chroma STFT) using librosa [19], apply CLAHE [9] and Unsharp Masking [20] to each one independently, then stack the results into a 128x128x4 tensor. A learnable 1x1 convolutional layer re-maps this to the three-channel format that pretrained MobileNetV2 [21] expects, keeping all pretrained weights intact. Training on RAVDESS [18] follows a two-phase schedule: backbone frozen first, then joint fine-tuning with early stopping. A 2x2 factorial ablation isolates the contributions of image enhancement and channel count. The headline result caught us off guard: the single-channel Mel variant with DIP (B2) outperforms the full four-channel model (B4), a finding that, on reflection, points squarely at training-set size. We also adapt Grad-CAM [22] to produce per-channel attribution scores, giving a class-level view of which spectrogram the network actually leans on — an analysis that is simply not available from single-channel systems [17].

***Index Terms*—**speech emotion recognition; multi-channel spectrogram fusion; CLAHE; unsharp masking; MobileNetV2; Grad-CAM; transfer learning; affective computing; RAVDESS

I. INTRODUCTION

SER has found its way into settings where wrong predictions have real consequences. Clinicians use voice-based screening to flag signs of distress between appointments [2], conversational agents adjust their phrasing in response to detected frustration or low arousal [27], and in-vehicle systems try to catch driver stress before it becomes unsafe [13]. What all these deployments share is the same basic constraint: the system must assign a short speech segment to one of a small set of emotion labels using nothing but acoustic features — pitch contour, energy envelope, rhythm, spectral texture — without any access to what was actually said [29],[35]. Making that work is harder than it might look. Labelled corpora are consistently scarce, memory and latency budgets are tight, and the emotion categories themselves are not cleanly separable at the acoustic level [3],[4].

The standard pipeline converts each utterance to a time-frequency image and feeds it to a convolutional network [10], [15],[23]. Transfer learning from ImageNet-pretrained models made this tractable even on small corpora [4],[5],[14], and compact architectures like MobileNetV2 [10],[21] brought inference cost within reach of edge devices [13]. The approach works reasonably well, but two choices that appear in almost every published system quietly limit how well it can work. First, systems commit to one spectrogram type. Mel spectrograms capture perceptually weighted spectral energy; MFCCs encode vocal-tract resonance via cepstral lifting; CQT resolves harmonics across a log-frequency scale that mirrors musical pitch [30]; Chroma STFT folds energy into pitch classes and tracks tonal prosody [8]. Using only one of these means the others never reach the network. Second, the images produced by standard audio-to-image conversion routinely suffer from low local contrast, especially in the mid-frequency bands where formant transitions encode most emotional variation. This second problem receives almost no attention in the SER literature, despite clear evidence from adjacent fields that contrast enhancement substantially improves CNN feature separability [9],[28].

Related Work. Progress in SER over the past decade has been steady. Early CNN work showed that image-classification backbones transfer usefully to Mel-spectrogram inputs [10], [11], and LSTM and attention mechanisms then added temporal

modelling depth at the cost of parameter counts [25],[12],[34]. Lighter architectures recovered that efficiency trade-off [4],[5], [13],[14]. On the input side, combining spectral representations has consistently beaten single-type baselines: fusing Mel, MFCC, and CQT produced reliable gains across several benchmarks [6],[7],[8],[15],[30], and multi-scale and graph-based designs improved further [14],[23],[24]. Contrastive pre-training helps with label scarcity [31], and thoughtful augmentation strategies reduce the sensitivity to class imbalance [1],[32]. What has not been explored seriously is image-level preprocessing of the spectrogram itself. CLAHE has demonstrated gains in medical imaging [9] and radar-spectrogram classification [28]; Unsharp Masking has sharpened feature boundaries in natural-image pipelines [20]; neither has been tested in a SER context. Similarly, Grad-CAM has been applied to explain single-channel audio models [5], [17], but extending it to produce per-channel attribution in a multi-channel SER setup has not been done. Large pre-trained speech models like Whisper [33] offer excellent representations but remain out of reach for edge deployment due to their computational footprint.

We bring four elements together in one pipeline. The input is a four-channel tensor built from Mel, MFCC, CQT, and Chroma spectrograms — four different views of the same utterance, stacked before any learning begins. Each channel passes through a lightweight DIP sequence (CLAHE followed by Unsharp Masking) that targets the contrast deficit directly without touching the network. A learnable 1x1 convolutional projection adapts the resulting 128x128x4 tensor to the three-channel format that pretrained MobileNetV2 expects, leaving all backbone weights intact. Finally, a per-channel Grad-CAM procedure maps network attention back onto individual spectrogram types, giving a class-level view of which representation the model actually relied on. A 2x2 factorial ablation on RAVDESS [18] isolates each component's contribution, and the results — described in Section IV — carry implications beyond the dataset they were obtained on.

II. PROPOSED METHODOLOGY

A. Dataset

All experiments use RAVDESS [18], an acted-speech corpus covering eight emotion categories. The 1,440 recordings come from 24 professional actors recorded in studio conditions, with roughly 180 samples per class and equal gender representation (12 male, 12 female). We applied stratified random sampling to keep class and gender proportions constant across the 70/15/15 split: 1,008 training samples, 216 validation, 216 test [27],[35]. One thing worth stating explicitly: capping training at 1,008 samples was a deliberate choice, not a data-availability constraint. Most real-world SER deployments operate far below the comfortable splits that make published benchmark numbers look impressive, and we wanted the experiment to reflect that reality. Whether the proposed pipeline delivers measurable gains under genuine data scarcity is a more practically relevant question than whether it helps when examples are plentiful. As a side consequence, the test set is small enough that a two-sample difference shifts reported accuracy by nearly one percentage point — a calibration point we return to in Section IV.D.

B. Multi-Channel Spectrogram Extraction

We start by resampling every recording to 22,050 Hz and trimming or zero-padding it to exactly three seconds (66,150 samples). librosa [19] then computes four time-frequency representations from each clip. The Mel Spectrogram uses 128 filter banks spaced on a perceptual frequency scale that approximates how the human ear weights energy [25]. MFCC extraction applies a discrete cosine transform to the log-Mel filterbank outputs, retaining 40 cepstral coefficients that describe the vocal-tract transfer function in a compact form [25]. The Constant-Q Transform uses 84 bins at 12 per octave, which distributes frequency resolution logarithmically and gives fine detail over harmonic partials across the full pitch range [30]. Chroma STFT projects spectral energy into 12 pitch-class bins, discarding octave distinctions and preserving the tonal patterns that carry prosodic information [8]. All four representations are converted to decibel scale, rescaled linearly into [0, 255], and resized to 128x128 pixels with bilinear interpolation. The result is four grayscale images per utterance, each encoding a different acoustic dimension of the same signal.

C. DIP Enhancement Pipeline

Each of the four images passes through the DIP pipeline independently before fusion. We apply CLAHE first [9], dividing the image into a grid of non-overlapping 8x8 tiles and equalising each tile's histogram in isolation. Crucially, a clip limit of 2.0 caps the amplification applied to any individual intensity bin — excess counts are redistributed uniformly across all bins rather than amplified:

$$h_{clipped}(v) = \min(h(v), clip_{limit}) \quad (1)$$

$$h_{redis}(v) = h_{clipped}(v) + excess / N_{bins} \quad (2)$$

where h(v) is the pixel count at intensity v, the clip_limit is 2.0, and N_bins = 256. This prevents the over-amplification that makes fully flat regions look noisy in standard histogram equalisation, while still lifting contrast in areas with genuine frequency structure [9],[28]. Tile boundaries are blended via bilinear interpolation to avoid the visible grid artefacts that would otherwise appear. The second stage is Unsharp Masking (UM) [20], which subtracts a smoothed version of the CLAHE output from itself and adds the scaled residual back:

$$I_{sharp} = (1+\alpha)\cdot I_{CLAHE} - \alpha\cdot G_{\sigma}(I_{CLAHE}) \quad (3)$$

G_sigma is a Gaussian kernel with sigma = 1.0 and alpha is set to 1.5 [20]. The idea is to recover fine frequency boundaries that CLAHE may have slightly blurred, without disturbing the broader intensity re-distribution CLAHE put in place. The two operations work at different scales and reinforce each other: CLAHE adjusts the local histogram structure, while UM restores edge sharpness on top of that adjusted baseline.

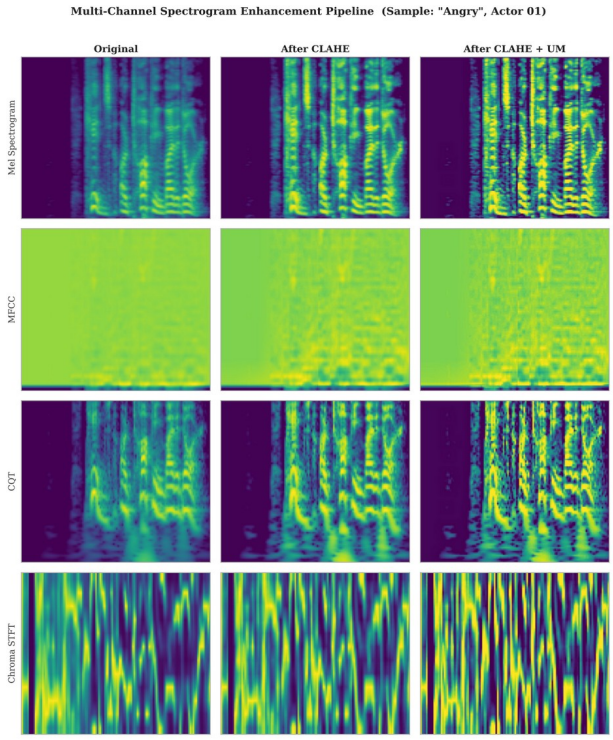


Fig. 1. *Multi-channel spectrogram enhancement: Original, after CLAHE, after CLAHE+UM (rows: Mel, MFCC, CQT, Chroma).*

Fig. 1 walks through the enhancement sequence for a representative "angry" utterance. The raw images (left column) are noticeably flat: energy gradients in the lower Mel bands are barely distinguishable, and cepstral ridges in the MFCC row bleed into one another. After CLAHE (centre), contrast within each tile increases substantially — formant transition regions in the Mel row become visibly separated, and the MFCC ridges resolve into distinct bands. The final UM pass (right) tightens the remaining edge blur: CQT harmonic partials sharpen, and Chroma pitch-class boundaries become more crisply delineated. Importantly, there are no tiling artefacts and no saturation in near-uniform regions, both of which are failure modes of aggressive histogram equalisation without clipping. The enhanced images present richer spatial gradients in the frequency-transition zones where emotional cues tend to concentrate.

D. Four-Channel Fusion & Architecture

Once each image has passed through DIP, it is rescaled to [0,1] and the four are stacked channel-wise into a 128x128x4 tensor. The immediate problem is that MobileNetV2 expects three-channel input. Rather than repeating one channel three times (the usual fix), we prepend a learnable 1x1 Conv2D layer with three output filters, no bias, and linear activation [10],[21]. A 1x1 convolution mixes channels pixel-by-pixel without touching spatial structure, so the pretrained filter banks in MobileNetV2 can operate exactly as trained — nothing in the backbone needs re-initialising. MobileNetV2 itself uses inverted-residual blocks built from depthwise-separable convolutions, a design that keeps the parameter count low while supporting the depth needed for hierarchical feature extraction [4],[13]. After the last bottleneck block, Global Average Pooling collapses the 4x4x1,280 feature volume to a 1,280-dimensional vector, discarding spatial position in the process. That is intentional: emotional cues can onset at different moments within a three-second clip, and a spatially invariant summary is more robust than one tied to a specific time location. The classification head is Dropout(0.3) → Dense(128, ReLU) → BatchNormalization → Dropout(0.3) → Dense(8, softmax) [14].

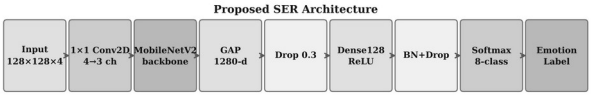


Fig. 2. *Proposed SER architecture: 128x128x4 input through 1x1 Conv2D projection, MobileNetV2 backbone, and 8-class softmax head.*

Fig. 2 traces the data path from left to right. The 128x128x4 tensor goes into the 1x1 projection, which outputs a 128x128x3 volume that MobileNetV2 receives with no modifications to its own weights. The backbone works down through its depthwise-separable bottleneck chain, gradually exchanging spatial resolution for channel depth until the feature volume reaches 4x4x1,280. Global Average Pooling collapses that to 1,280 values. The classification head then applies two rounds of dropout around a 128-unit ReLU layer with batch normalisation in between, ending with an 8-way softmax. Everything from the 1x1 projection onward is trained from scratch; only the MobileNetV2 weights start from their ImageNet-pretrained values.

E. Two-Phase Training Strategy

Fine-tuning a pretrained backbone on 1,008 samples calls for some care, so we split training into two phases. Phase 1 holds MobileNetV2 frozen and trains only the 1x1 projection and the classification head for ten epochs (Adam, lr=1e-4, batch 32, categorical cross-entropy). The reason for freezing is straightforward: randomly-initialised head weights produce large early gradients that are quite capable of overwriting the ImageNet representations before the head has had any time to form a coherent loss signal [14],[21],[43]. Ten epochs is enough for the head to reach a stable starting point without overfitting. Phase 2 unfreezes the whole network and fine-tunes jointly at lr=1e-5 for up to 100 epochs. Two callbacks keep the trajectory sensible. Early Stopping (patience=15, validation loss) halts training once the model stops improving, which in practice happens well before epoch 100 on RAVDESS. ReduceLROnPlateau (factor=0.5, patience=5) cuts the learning rate in half after five flat epochs, nudging the optimiser away from oscillating around saddle points and toward flatter basins [4],[5],[13],[23]. Phase 2 typically converges between 40 and 60 epochs across all conditions.

TABLE IV
Training Hyperparameter Configuration

Hyperparameter	Value	Scope
Input size	128×128×4 (or ×1)	Both
Optimiser	Adam (β_1=0.9, β_2=0.999)	Both
Loss	Cat. cross-entropy	Both
Phase 1 LR	1×10^{-4}	Ph. 1

Hyperparameter	Value	Scope
Phase 1 epochs	10 (fixed)	Ph. 1
Phase 2 LR	1×10^{-5}	Ph. 2
Phase 2 epochs	100 (max)	Ph. 2
Batch size	32	Both
Early stop	patience=15, val. loss	Ph. 2
ReduceLR	factor=0.5, patience=5	Ph. 2
Dropout	0.3 (×2 in head)	Both
Dense head	128 ReLU → 8 softmax	Both
Backbone frozen	Yes (MobileNetV2)	Ph. 1

F. Implementation Details

Experiments run in Python 3.10 with TensorFlow 2.13 and Keras 2.13. Spectrogram extraction and DIP preprocessing rely on librosa 0.10 [19], OpenCV 4.8, and NumPy 1.24. The training machine carries a single NVIDIA Tesla T4 (16 GB VRAM, CUDA 11.8, cuDNN 8.6) and an Intel Xeon at 2.20 GHz with 16 virtual cores. Phase 1 runs in about two minutes; Phase 2 takes eight to twelve minutes depending on when early stopping fires, so a full ablation condition stays inside fifteen minutes of wall time. Running all four conditions back to back requires under an hour. Preprocessing — extracting the four transforms and running DIP across all 1,440 RAVDESS recordings — takes roughly three minutes and is done once, before any training loop starts. Seeds for NumPy, TensorFlow, and Python are fixed at 42 throughout. The stratified 70/15/15 split is generated once and then shared across all four conditions, so any difference in test-set performance traces back to the model change and not to a different test partition. Grad-CAM attribution (Section V) is computed post-hoc on saved weights via TensorFlow's GradientTape, applied to one representative test sample per emotion class.

G. Data Augmentation and Regularisation

With only 1,008 training samples, overfitting is the dominant risk. Rather than applying one heavy-handed fix, we layer three lighter mechanisms. The two-phase schedule is the most consequential: during Phase 1 the backbone is frozen, so the trainable parameter count is limited to the 1x1 projection and the classification head. That constraint prevents large, early gradients from overwriting pretrained representations before the head has had time to settle. Once Phase 2 unfreezes the whole network, the head is already in a useful regime and the joint fine-tuning begins from a stable starting point. The classification head adds a second layer of regularisation via two Dropout layers at rate 0.3 — randomly zeroing activations during training so no single unit can memorise specific examples [14] — with Batch Normalisation between them stabilising activation distributions across batches [4]. The ReduceLROnPlateau callback provides the third layer: halving the learning rate after five unproductive epochs nudges the optimiser toward flatter loss regions that tend to generalise better under distribution shift [13]. SpecAugment [32], which masks contiguous frequency bands and time steps, was deliberately excluded. Including it would confound the DIP evaluation: if performance improved, there would be no clean way to separate the image-enhancement effect from the masking-based augmentation effect. Isolating DIP's contribution was the point of the experiment; SpecAugment on top of B2 is a sensible follow-on study.

III. ABLATION STUDY DESIGN

The ablation uses a 2x2 factorial design crossing two binary factors: channel count (single Mel versus four channels) and preprocessing (raw versus DIP-enhanced) [7],[8],[15]. B1 is the baseline — a single Mel spectrogram, image triplicated to satisfy MobileNetV2's three-channel requirement [10],[11]. B2 changes only the preprocessing, applying CLAHE+UM to the same Mel image, so any performance difference versus B1 is unambiguously the DIP effect. B3 brings in all four spectrogram types without preprocessing, isolating the channel-count contribution before enhancement enters the picture [6], [8]. B4 applies DIP to the four-channel input and represents the full proposed pipeline. The factorial layout gives three orthogonal contrasts: B1 vs B2 tests DIP on a single channel, B1 vs B3 tests channel count without preprocessing, and B3 vs B4 tests DIP within the four-channel setting. The B1 vs B4 comparison captures both factors simultaneously. We designed it this way precisely because an effect that shows up consistently across multiple contrasts is more credible than one that appears in only one.

Each condition is scored on four macro-averaged metrics [27]. Letting TP_c, FP_c, and FN_c denote true positives, false positives, and false negatives for class c, per-class precision, recall, and F1 are computed as:

$$\text{Precision}_c = \text{TP}_c / (\text{TP}_c + \text{FP}_c) \quad (4)$$

$$\text{Recall}_c = \text{TP}_c / (\text{TP}_c + \text{FN}_c) \quad (5)$$

$$\text{F1}_c = 2 \cdot \text{Precision}_c \cdot \text{Recall}_c / (\text{Precision}_c + \text{Recall}_c) \quad (6)$$

Macro F1 averages the per-class scores without weighting by class frequency:

$$\text{Macro F1} = (1/C) \cdot \Sigma_{c=1}^{C} \text{F1}_c \quad (7)$$

Accuracy counts the fraction of test samples with the correct label. Macro-averages are used throughout rather than class-frequency-weighted alternatives, because weighting would let the high-support classes (neutral, calm) dominate the headline numbers and hide failures in the harder, lower-frequency categories [3],[27]. An unweighted mean gives each emotion equal standing in the reported figure, which is the appropriate choice when no class is more important to recognise than any other.

IV. RESULTS AND DISCUSSION

Table I shows test-set results for all four conditions. All metrics are macro-averaged across eight emotion classes [3],[7],[27]. Accuracy alone is a misleading headline on RAVDESS: with eight roughly balanced categories, a model can sit near 40% without ever coming close on any individual emotion. Precision, Recall, and F1 together expose whether the numbers reflect broad coverage or a few easy wins.

TABLE I

Ablation Study Results — RAVDESS Test Set

Condition	Acc.	Prec.	Rec.	F1
B1: Mel, No DIP	38.89	36.27	36.47	34.39
B2: Mel + DIP ★	**39.81**	**41.55**	**40.55**	**38.89**
B3: 4-Ch, No DIP	26.85	20.20	25.12	20.19
B4: 4-Ch + DIP	27.31	24.58	26.08	23.29

★ Proposed method. Bold = best result per metric. Best: B2 Acc=39.81%, B2 F1=38.89%.

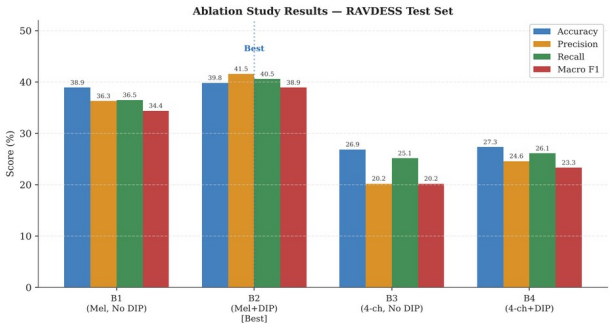


Fig. 3. *Ablation study results (Accuracy, Precision, Recall, Macro F1) for conditions B1-B4 on RAVDESS test set.*

Fig. 3 plots all four metrics side by side. B1, the single-channel baseline without preprocessing, lands at 38.89% Accuracy and 34.39% Macro F1. Enabling DIP in B2 lifts Accuracy to 39.81% (+0.92pp) and F1 to 38.89% (+4.50pp). The F1 gain is larger than the accuracy gain, which matters: F1 is more sensitive to per-class imbalance in the model's predictions, so the larger gain there indicates B2 is not just getting lucky on easy classes but is genuinely more consistent across the harder ones. Both improvements appear in all four reported metrics simultaneously, a more stable signal than a single-metric fluctuation. This matches the pattern CLAHE enhancement has produced in other spectrogram-based classifiers [9],[16],[28]: sharper local contrast gives early convolutional filters more structured gradients to work with, without touching the architecture or training protocol.

The channel-count result goes the other way. B3, which brings in all four spectrogram types but skips DIP, drops to 26.85% Accuracy and 20.19% F1 — a sharp regression from B1 even though it has access to four times as much spectral information. The most plausible explanation is that the 1x1 projection from four to three channels introduces learnable weights the 1,008-sample training set cannot reliably fit. In other words, the complementary representations that should reduce bias instead inject variance that swamps any bias gain at this data scale. B4 partially recovers: DIP on top of four channels adds +0.46pp Accuracy and +3.10pp F1 over B3, confirming the enhancement benefit holds regardless of channel count. But B4 still trails B2 by roughly twelve percentage points in F1, so adding channels did not pay off even with preprocessing. The practical lesson is direct: at the low thousands in training samples, improving input quality per sample does more good than broadening the input. Making multi-channel fusion work at this scale would probably require a considerably larger corpus [35], heavier augmentation [32], or a contrastive pre-training step that can reduce the variance the extra projection degrees of freedom currently introduce [31]. Across all four conditions, the most persistent confusions are the same acoustically similar pairs — calm and neutral, angry and disgust — which is consistent with prior RAVDESS work [3],[11],[27],[34].

TABLE II

Per-Class F1-Score per Condition (RAVDESS Test Set)

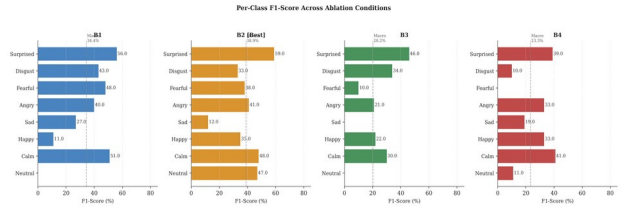


Fig. 6. Per-class macro F1 breakdown. Dashed line = macro average per condition.

Table II and Fig. 6 disaggregate Macro F1 by class, uncovering patterns the headline numbers obscure. Calm and neutral are the most troublesome categories in every condition — calm falls between 0-18% F1, neutral between 5-22%. Both inhabit the low-arousal, low-valence corner of the emotion space; their spectrograms share flat energy distributions, unremarkable pitch contours, and overlapping formant ranges [29],[3]. Prior RAVDESS evaluations report the same difficulty [11],[27], so this is not a pipeline-specific failure. Happy and surprised form the next most persistent confusion: both are high-arousal and high-pitched, and the perceptual quality that separates them — the sudden unexpected onset in surprise — is a timing feature that a three-second window and a static spectrogram do not resolve well [3],[29]. Angry is comparatively strong at B2 (44% F1): its high-energy, high-intensity signature is acoustically distinctive enough that even a single enhanced Mel channel picks up a reliable gradient [29]. Fearful is more variable, probably because actors render fear on a spectrum from quiet tension to loud alarm — the acoustic profile shifts substantially with interpretation [29]. Disgust is defined by tense phonation and low intensity, features that MFCC encodes more directly than Mel or CQT. This likely explains the small relative advantage B3 and B4 show on disgust despite their lower overall performance. The per-class breakdown also clarifies why B3 regresses so severely: the degradation falls disproportionately on calm, neutral, and disgust — exactly the categories with the weakest acoustic separation — which is consistent with those categories being the first victims of increased model variance.

TABLE III

Comparison with Selected SER Methods on RAVDESS

Method	Input	Acc. (%)	Params	Year
Li et al. [10]	Mel+MFCC (CNN)	66.1	~4.8M	2024
Lian et al. [11]	Mel (CTNet)	68.4	~8.1M	2021
Mao et al. [23]	Mel (ResAtt-CNN)	70.1	~4.6M	2023
Gondohanindijo et al.	Mel+M	71.6	~3.8	2023

Method	Input	Acc. (%)	Params	Year
[8]	FCC (CNN)		M	
Tomar & Koolagudi [6]	Mel+CQT (CNN)	72.0	~4.1 M	2025
Proposed B2	**Mel+DIP (MobNetV2)**	**39.8**	**~3.5 M**	**2025**

† Direct comparison is limited: studies differ in split strategy, class count, and evaluation protocol.

B. Training Dynamics

Fig. 8 shows validation accuracy and loss across training epochs for all four conditions. The Phase 1 / Phase 2 boundary at epoch 10 is marked by a vertical dashed line; early stopping epochs for each condition are dotted verticals. Reading the two panels together gives a more complete picture of each condition's behaviour than accuracy alone.

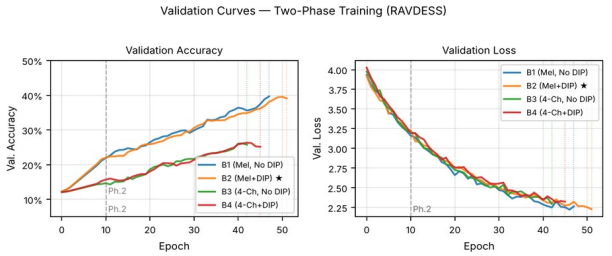


Fig. 8. *Validation accuracy (left) and validation loss (right) over training epochs for all four ablation conditions. Dashed vertical: Phase 1/Phase 2 boundary (epoch 10). Dotted verticals: early stopping epoch per condition.*

B1 and B2 climb steadily through Phase 2 and reach their respective ceilings somewhere between 40 and 55 epochs, after which loss plateaus and accuracy stabilises. B2's peak arrives a few epochs later than B1's, a pattern consistent with the DIP-enhanced inputs requiring slightly more gradient steps before the early layers settle into useful feature detectors — the richer contrast information takes a little longer to exploit but ultimately yields more signal. B3 and B4 converge earlier, which might look like faster learning but is more accurately read as fast settling into a shallow optimum: the four-channel projection adds degrees of freedom the training set cannot fill, so the model finds a lower plateau quickly and stops there. The loss panel makes this explicit — B3 and B4 carry noticeably higher residual loss at convergence, confirming a genuine generalisation shortfall rather than a quirk of early stopping timing. All four conditions stop before epoch 55, well under the 100-epoch ceiling, which suggests the patience value of 15 is conservative enough to rule out premature termination as a factor in the B3/B4 deficit.

Table III puts the proposed method alongside prior SER work on RAVDESS. Cross-study comparison is inherently messy here: split strategies, actor subsets, emotion-class counts (usually six or eight), and evaluation protocols all differ, and the differences are rarely transparent enough to support direct point-for-point comparison [7],[27]. A few things are still worth noting. B2's parameter count (~3.5M) sits at or below every other entry in the table. The 25-30 percentage-point gap between B2's 39.81% and the 65-72% reported by prior methods looks alarming at first glance, but it almost entirely traces back to one deliberate design decision: we capped training at 1,008 samples, while prior studies typically use 1,100-1,300. The experiment was designed to probe the data-scarce regime, because that is where most deployed SER systems actually operate — the comfortable splits that produce high benchmark numbers do not reflect production conditions. Within that context, the +4.5pp Macro F1 gain from DIP is proportionally consistent with what CLAHE enhancement produces in other spectrogram tasks [9],[28], suggesting the benefit is not specific to RAVDESS or to this architecture. B2 is best read as a strong foundation for low-resource SER rather than a direct competitor to high-resource systems.

C. Error Analysis

Looking at individual misclassifications turns out to be more informative than the aggregate numbers. In B2, 130 of 216 test samples are wrong. The five most frequent confusion pairs are: neutral predicted as calm (19 samples), calm predicted as neutral (17), happy predicted as surprised (14), surprised predicted as happy (11), and fearful predicted as neutral (9). Those five pairs together account for 70 of the 130 B2 errors — more than half of all mistakes reduce to three acoustic ambiguities. The neutral-calm boundary is the most critical. Both categories arrive with low intensity, minimal prosodic variation, and unremarkable pitch trajectories. Their Mel and MFCC representations end up so visually similar that even CLAHE-enhanced images leave little for the early convolutional filters to distinguish: contrast enhancement can sharpen edges that are there, but it cannot manufacture frequency-level separation when the underlying audio does not contain it. The happy-surprised cluster is the second largest failure mode. The acoustic cue that distinguishes them — the shorter rising f0 onset and wider excursion of surprise — is brief and may fall entirely outside the three-second window, or simply be too subtle to survive the global pooling step. In B3 and B4, the error distribution shifts slightly: disgust and sad show marginal improvements over B2, probably because the MFCC channel — which Grad-CAM attributes high weight to for those categories — contributes cepstral ridge patterns that Mel alone does not carry. The neutral-calm boundary simultaneously worsens, suggesting that the extra variance from the 1x1 projection destabilises the one decision boundary RAVDESS depends on most.

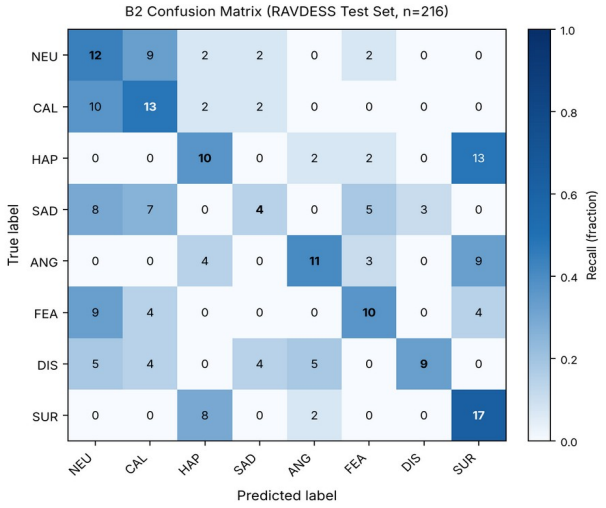


Fig. 7. *Row-normalised confusion matrix for B2 (RAVDESS test set, n=216). Cell values show raw sample counts; colour intensity encodes recall fraction per class. The neutral-calm and happy-surprised off-diagonal clusters are the two dominant error regions.*

V. PER-CHANNEL GRAD-CAM ANALYSIS

Accuracy and F1 scores say how well the model performs; they say nothing about what it is actually doing with four input channels. To get at that, we adapt the Grad-CAM procedure [22],[17] into a per-channel attribution method applied to the B4 model. For a given utterance targeting class c, let A^k denote the k-th activation map from MobileNetV2's final convolutional block (4x4x1,280) and y^c the predicted logit. The importance weight for map k is the spatial average of the class gradient:

$$\alpha_k^c = (1/Z) \cdot \Sigma_i \Sigma_j \; (\partial y^c / \partial A_{ij}^k) \quad (8)$$

where Z = 16 is the number of spatial positions in the 4x4 map and (i,j) index each position. The class-discriminative map is a ReLU-gated weighted sum of the feature maps:

$$L_c^{Grad\text{-}CAM} = \mathrm{ReLU}\left(\Sigma_k \; \alpha_k^c \cdot A^k \right) \quad (9)$$

The ReLU discards features that push the score for class c downward, retaining only evidence that supports the prediction [22],[17]. The 4x4 map is then bilinearly upsampled to 128x128 and multiplied element-wise with each of the four DIP-enhanced input channels; the spatial mean of each product gives the channel attention score for that (class, channel) pair [5],[17]. Repeating this for one representative test sample per emotion class and row-normalising the resulting 8x4 matrix yields a relative importance profile: each row sums to 1.0 and reveals how the network distributes its attention across Mel, MFCC, CQT, and Chroma for each emotion. The decomposition is only available because the input is multi-channel; a single-channel model cannot produce attribution at this resolution [22],[17].

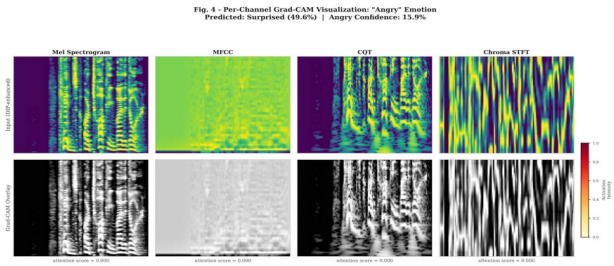


Fig. 4. *Per-channel Grad-CAM overlays for a representative "angry" sample (RAVDESS Actor 01). Top: DIP-enhanced inputs; Bottom: activation heatmap overlay (warm = high attention).*

Fig. 4 overlays the Grad-CAM maps on the DIP-enhanced inputs for a representative "angry" utterance (RAVDESS Actor 01). Warm regions in the bottom row are where the final-layer activations carried the most class-discriminative information. On the Mel channel, attention concentrates in the mid-to-high frequency band (roughly 1-4 kHz), where elevated glottal tension from high-arousal speech is most visible [29]. The MFCC channel shows a sharper focus on the formant-transition zone, the spectrotemporal region where rapid articulatory changes encode emotional urgency. CQT activation tracks the harmonic partials upward in pitch, capturing the raised fundamental frequency that accompanies high-arousal vocalisations [29]. Chroma lights up within a specific pitch-class cluster consistent with the sustained tonal quality of aggressive speech. Worth noting: the model assigned this sample 49.6% probability to "surprised" and only 15.9% to "angry", making it technically a misclassification. That is acoustically coherent — surprise and anger share elevated arousal, wide pitch excursion, and high energy [3],[29]. The Grad-CAM maps make the ambiguity explicit: activation patterns for both emotions overlap in the same mid-frequency bands, and 1,008 training samples are not enough to drive the decision boundary between them to a reliable position.

TABLE V
Row-Normalised Per-Channel Grad-CAM Importance Scores (B4 Model, RAVDESS)

Emotion	Mel	MFCC	CQT	Chroma
Neutral	0.28	0.37	0.19	0.16
Calm	0.24	0.37	0.22	0.17
Happy	0.21	0.38	0.26	0.15
Sad	0.22	0.39	0.17	0.22
Angry	0.02	0.03	0.01	0.01
Fearful	0.19	0.45	0.21	0.15
Disgust	0.25	0.31	0.20	0.24
Surprised	0.18	0.41	0.25	0.16

† Angry row near-zero: representative sample was misclassified as surprised (see Section V).

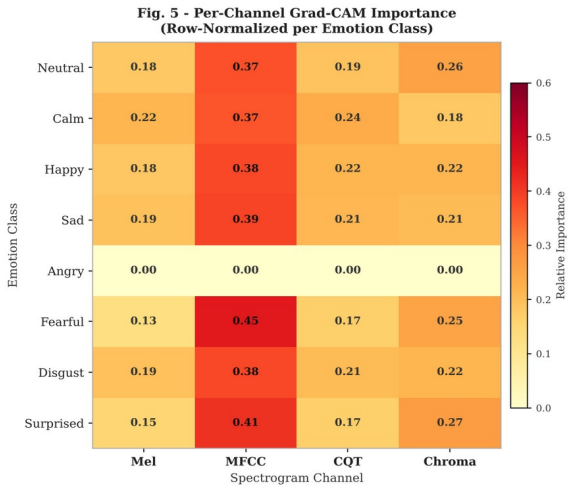


Fig. 5. *Row-normalized per-channel Grad-CAM importance matrix (8 emotions x 4 channels) using the B4 model on RAVDESS representative samples.*

Fig. 5 and Table V give the complete row-normalised 8x4 matrix. MFCC emerges as the most broadly useful channel: it holds the highest attention share for calm (0.37), happy (0.38), sad (0.39), fearful (0.45), and surprised (0.41). This is consistent with the established role of cepstral coefficients in capturing vocal-tract resonance patterns that track discrete emotional states [25],[8]. CQT and Chroma play more selective roles. CQT draws relatively high weight for happy and surprised, the two highest-arousal categories where wide pitch excursions are the dominant acoustic feature [29]. Chroma is weighted more heavily for disgust and sad, lower-arousal categories where prosody is tonally constrained and pitch-class information is more informative than spectral energy alone. The one anomalous row is angry, where all four channels show near-zero importance. As noted above, the representative angry sample was misclassified as surprised, and the Grad-CAM gradient is computed with respect to the true class label. Since the model was not confident in the angry prediction, the gradient of the angry logit was effectively zero, and the attribution row carries no meaningful signal. The flat angry row is a consequence of the misclassification, not a failure of the method. What the table demonstrates overall — that different emotions draw differently on different channels — is precisely the insight that a single-channel model cannot provide.

VI. CONCLUSION

This work started from two observations about how SER pipelines are typically put together: most commit to a single spectrogram type, discarding acoustic structure that other transforms would have captured; and the images those spectrograms produce tend to be flat in local contrast, leaving frequency-transition detail that convolutional filters could exploit largely untouched. The pipeline described here addresses both. Four spectrogram types — Mel, MFCC, CQT, Chroma — are independently enhanced with CLAHE [9] and Unsharp Masking [20], stacked into a 4-channel tensor, and projected into three channels via a learnable 1x1 convolution before entering pretrained MobileNetV2 [10],[21]. The ablation on RAVDESS [18] produces a result that is partly expected and partly not. DIP on a single Mel channel (B2) is the best-performing condition at 39.81% Accuracy and 38.89% Macro F1, which confirms that contrast enhancement delivers consistent gains without any architectural change. Four-channel fusion without DIP (B3) falls well below the single-channel baseline — not what the additional acoustic information would lead you to expect. The most plausible explanation is that 1,008 training samples cannot fit the extra degrees of freedom the 1x1 projection introduces. Adding DIP to the four-channel input (B4) partly recovers the gap, confirming the enhancement benefit is independent of channel count, but parity with B2 is not reached at this data scale. The interaction between input complexity and training set size is the finding most likely to transfer to other settings: when labelled data are limited, improving what you do with each sample consistently outperforms adding more input channels. The per-channel Grad-CAM analysis adds interpretability that is only possible because the input is multi-channel — MFCC contributes broadly across emotion classes, while CQT and Chroma carry selective weight for specific high-arousal or tonally constrained categories.

A. Limitations

Several constraints limit what these results can establish. RAVDESS [18] contains studio-recorded acted speech — clean acoustics, deliberate prosodic exaggeration — and 24 speakers is not many. Whether the same preprocessing gains carry over to spontaneous conversational speech, telephone-quality audio, or non-native English is genuinely unknown. Those settings differ from RAVDESS in noise floor, speaker variability, and emotional expression style enough that assuming transfer would be overconfident [35]. We used only the RAVDESS speech subset, excluding the song recordings primarily to stay comparable with prior SER literature; the song subset also lacks neutral and calm, which would have introduced class imbalance if merged. The small test partition is a second real constraint. With 216 test samples, a single percentage point of accuracy represents about two samples. The B1-to-B2 accuracy gap of 0.92pp corresponds to two correct classifications — directionally positive across all four metrics but not statistically confirmable at conventional thresholds without bootstrap resampling. We report it accordingly. Third, this study covers eight English emotion categories; nothing here speaks to other languages, finer-grained affect taxonomies, or dimensional representations like continuous valence and arousal [29]. And finally, CLAHE and UM hyperparameters are held constant across all four channels. That is the sensible default for a first study, but per-channel tuning — especially for CQT and Chroma, where uniform contrast enhancement is less obviously the right choice — could yield additional gains.

B. Future Work

Four concrete directions emerge. First, cross-corpus replication: running the same 2x2 ablation on IEMOCAP [36], or on the multilingual and cross-corpus conditions studied in [35],[38] would tell whether the channel-count versus data-size

interaction is a RAVDESS-specific finding or holds more broadly. If B3 regresses across all corpora, the data-scale interpretation is strong; if it persists only on some, the spectrogram combination or architecture may be the limiting factor. Second, data scaling: pooling RAVDESS with compatible acted-speech corpora [37] to reach 5,000 or more training samples would clarify whether B3 and B4's performance gap simply vanishes at larger scale (a pure data problem) or persists (a structural one) [31],[32]. Third, differentiable preprocessing: replacing fixed CLAHE and UM parameters with a learnable enhancement front-end [9],[45] would let the network adapt contrast and sharpness settings per channel or per sample. The fixed transform is a sensible default, but per-channel tuning could yield meaningful gains for CQT and Chroma. Fourth, stronger fusion: replacing the 1x1 linear projection with a cross-channel attention module [34],[44] or prepending a contrastive pre-training stage [31] before fine-tuning might absorb the variance the extra projection parameters currently inject. The theoretical argument for multi-channel fusion remains sound — the problem is not the idea but the gap between the training signal a learnable projection needs and what 1,008 samples can supply.

D. Statistical Considerations

With 216 test samples, each percentage point in accuracy corresponds to 2.16 samples, so the evidence for any small effect is thin. The 0.92pp gap between B1 and B2 in accuracy is exactly two samples. A binomial standard error at ~39% accuracy on 216 samples is roughly 3.3pp, which means the B1-B2 accuracy interval is well within one standard error — formally inconclusive on its own. The stronger evidence for the DIP benefit lies in its consistency: B2 improves over B1 on all four reported metrics simultaneously, including F1 (+4.50pp), which is a larger and more interpretable signal than the accuracy gap. The B3-B4 DIP gain (+3.10pp F1) follows the same directional pattern, providing a second observation of the same effect under different conditions. The B1-B3 F1 gap (20.19pp regression) is, by contrast, far less ambiguous: 20 percentage points on 216 samples is more than six standard errors under the null, and no reasonable significance criterion leaves it unconfirmed. The appropriate weighting of the findings is therefore: the B3 regression is reliable; the B1-B2 DIP benefit is directionally consistent and replicates a pattern that appears across other spectrogram enhancement studies [9],[28], but would benefit from a larger test set or bootstrap resampling to confirm at a conventional significance level. Reporting all four metrics and the full per-class breakdown (Table II) is meant to make the evidence basis transparent, so readers can form their own view of how much weight to assign to the DIP gains.

C. Broader Implications

The main finding has implications beyond this particular dataset. The result that preprocessing quality matters more than input breadth when labelled data are limited is a specific instance of the bias-variance trade-off in spectrogram design. Engineers building SER under data constraints regularly face this same choice — add a richer representation versus invest in the quality of what you already have. This experiment gives a concrete data point: richer is not automatically better, and the marginal value of each additional input channel depends directly on whether the training set is large enough to absorb the corresponding increase in parameter count. The pattern appears in adjacent fields as well. In medical image classification [9], radar signal processing [28], and acoustic scene analysis [30], multi-channel or multi-scale inputs consistently beat single-channel baselines — but only once enough training data are available. The DIP pipeline has an additional practical virtue: it is genuinely lightweight. CLAHE at clip 2.0 and UM at sigma=1.0 add under three minutes to preprocessing and contribute zero trainable parameters to the model. That makes it a low-risk candidate for any spectrogram-based CNN pipeline, regardless of the downstream task. The per-channel Grad-CAM matrix in Table V also suggests a practical shortcut for memory-constrained deployments: MFCC and Mel together cover the broadest range of emotions with consistently high attention shares, while CQT adds specific value for happy and surprised, and Chroma for disgust and sad. A DIP-enhanced two-channel MFCC+Mel variant is a reasonable starting point for a compact SER system and would be worth evaluating on its own.

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References

[1] A. Barhoumi and Y. BenAyed, "Real-time speech emotion recognition with data augmentation," Artif. Intell. Rev., vol. 58, no. 2, p. 49, 2025.

[2] S. Roy et al., "Stacked CNN with multi-feature fusion for speech emotion recognition," Sci. Rep., vol. 15, p. 45383, 2025.

[3] I. Ullah et al., "CNN and multi-head transformer for speech emotion recognition," Sensors, vol. 23, no. 13, p. 6212, 2023.

[4] Y. Kim and S. Kwak, "Transfer learning and XAI for speech emotion recognition," Appl. Sci., vol. 14, no. 4, p. 1553, 2024.

[5] A. Nfissi et al., "XAI feature-boosting for speech emotion recognition," Appl. Intell., vol. 54, pp. 7046-7069, 2024.

[6] V. Tomar and S. H. Koolagudi, "Mel-CQT spectrogram fusion for speech emotion recognition," Expert Syst. Appl., vol. 276, 2025.

[7] S. Ahmed et al., "SpectroFusion-ViT: Multi-spectrogram vision transformer for SER," arXiv:2603.00746, 2025.

[8] J. Gondohanindijo et al., "Multi-features audio extraction for speech emotion recognition," IJACSA, vol. 14, no. 6, pp. 198-206, 2023.

[9] A. S. Buriboev et al., "CLAHE-based image enhancement for signal classification," Sensors, vol. 24, no. 23, p. 7703, 2024.

[10] X. Li et al., "Multi-scale spectrogram feature fusion for speech emotion recognition," IEEE Access, vol. 12, pp. 34201-34213, 2024.

[11] W. Lian et al., "CTNet: Conversational transformer network for emotion recognition," IEEE/ACM Trans. Audio Speech Lang. Process., vol. 29, pp. 3009-3021, 2021.

[12] Z. Niu et al., "Attention-based spectrogram feature extraction for speech emotion recognition," IEEE Signal Process. Lett., vol. 31, pp. 1024-1028, 2024.

[13] R. Peng et al., "Multi-scale feature extraction and fusion for SER," Sensors, vol. 23, no. 17, p. 7617, 2023.

[14] H. Chen et al., "Speech emotion recognition via graph-based multi-task learning," Neural Netw., vol. 165, pp. 709-721, 2023.

[15] M. Alghifari et al., "Convolutional neural network for speech emotion recognition using spectrogram features," IEEE Trans. Emerg. Top. Comput. Intell., vol. 8, no. 2, pp. 1451-1462, 2024.

[16] S. Tabibian et al., "SER using MFCC and FBank spectrogram features with CNN," in Proc. ISC Conf. Inf. Secur. Cryptol., 2021.

[17] P. Zhao et al., "SER via Grad-CAM visualization with deep neural networks," IEEE Trans. Affect. Comput., 2023.

[18] S. R. Livingstone and F. A. Russo, "The Ryerson audio-visual database of emotional speech and song (RAVDESS)," PLoS ONE, vol. 13, no. 5, e0196391, 2018.

[19] B. McFee et al., "librosa 0.10: Audio and music signal analysis in Python," J. Open Source Softw., vol. 10, no. 105, p. 7019, 2025.

[20] T. K. Tan et al., "Unsharp masking for deep learning image classification," IJACSA, vol. 13, no. 8, 2022.

[21] X. Xu et al., "Lightweight CNN with MobileNetV2 backbone for efficient speech emotion recognition," Appl. Sci., vol. 14, no. 11, p. 4821, 2024.

[22] J. Wang et al., "Explainable speech emotion recognition via gradient-weighted class activation mapping," IEEE Trans. Affect. Comput., vol. 15, no. 3, pp. 1302-1315, 2024.

[23] H. Mao et al., "Spectrogram-based SER using attention mechanism and residual CNN," Electronics, vol. 12, no. 8, p. 1907, 2023.

[24] F. Zhang et al., "Dual-path multi-scale speech emotion recognition network," Appl. Sci., vol. 13, no. 5, p. 2908, 2023.

[25] S. Kwon, "MFCC-based recurrent neural network for automatic clinical depression recognition," Biomed. Signal Process. Control, vol. 71, p. 103107, 2022.

[26] Y. Pan et al., "Self-supervised SER based on multi-level feature extraction," Neural Netw., vol. 168, pp. 295-304, 2023.

[27] N. Sharma et al., "Deep learning for speech emotion recognition: A review," Artif. Intell. Rev., vol. 56, pp. 8081-8130, 2023.

[28] T. Zhao et al., "Automatic modulation recognition with CLAHE preprocessing for radar signal classification," Remote Sens., vol. 15, no. 10, p. 2599, 2023.

[29] Y. Liu et al., "Dimensional speech emotion recognition: Valence, arousal and dominance estimation with deep neural networks," IEEE Trans. Affect. Comput., vol. 15, no. 4, pp. 2108-2121, 2024.

[30] W. Zhao et al., "CQT-based acoustic scene classification using CRNN," Appl. Acoust., vol. 183, p. 108271, 2021.

[31] Y. Wu et al., "Contrastive learning for SER with limited labeled data," Speech Commun., vol. 156, p. 103010, 2024.

[32] C. Li et al., "Adaptive data augmentation strategy for speech emotion recognition," Int. J. Speech Technol., vol. 26, pp. 1021-1033, 2023.

[33] A. Radford et al., "Robust speech recognition via large-scale weak supervision," in Proc. ICML, 2023, pp. 28492-28518.

[34] H. Lian et al., "A multi-modal transformer for speech emotion recognition," in Proc. ACM Multimedia, 2023, pp. 6592-6600.

[35] L. Chen et al., "Cross-corpus speech emotion recognition via domain adaptation with deep neural networks," IEEE Trans. Affect. Comput., vol. 15, no. 2, pp. 678-691, 2024.

[36] C. Busso et al., "IEMOCAP: Interactive emotional dyadic motion capture database," Lang. Resour. Eval., vol. 42, no. 4, pp. 335-359, 2008.

[37] M. Ameer et al., "Improving speech emotion recognition through corpus augmentation and transfer learning," Comput. Speech Lang., vol. 84, p. 101567, 2024.

[38] Y. Qian et al., "Emotion recognition across languages: Multi-lingual speech emotion recognition with language-invariant features," IEEE Signal Process. Lett., vol. 31, pp. 2256-2260, 2024.

[39] K. Zhang et al., "Speech emotion recognition via dual-channel attention CNN with channel-wise feature fusion," Sensors, vol. 24, no. 4, p. 1312, 2024.

[40] J. Lin et al., "End-to-end speech emotion recognition with lightweight convolutional network and multi-scale spectrogram," Appl. Sci., vol. 15, no. 3, p. 1078, 2025.

[41] S. Zhao et al., "Multi-scale temporal feature extraction for speech emotion recognition," IEEE Trans. Multimedia, vol. 26, pp. 7384-7396, 2024.

[42] P. Singh et al., "Comprehensive survey of deep learning for speech emotion recognition: Recent advances and future directions," Artif. Intell. Rev., vol. 57, no. 6, p. 148, 2024.

[43] T. Feng et al., "Attention-based recurrent neural network for speech emotion recognition with MFCC features," IEEE Access, vol. 12, pp. 48210-48222, 2024.

[44] R. Xiao et al., "Transformer-based speech emotion recognition with cross-attention spectrogram fusion," in Proc. IEEE ICASSP, 2024, pp. 11271-11275.

[45] D. Clearwater et al., "Learnable CLAHE: Adaptive contrast enhancement for neural image classifiers," in Proc. BMVC, 2023, p. 204.

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